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General description

The electronic filling carrousel, type EFC/U-24, is a rotating platform designed for stationary installation on floor, with connection to a chain conveyor plant via automatic introduction and ejection units. The filling carrousel can accommodate up to 24 filling machines type UFM for filling of LPG cylinders.

Filling machines mounted on a carrousel that is connected to a chain conveyor plant is usually recommended as the most economical and rational way to fill LPG cylinders, if the desired or necessary filling capacity exceeds 250 domestic cylinders per hour.

By using a filling carrousel and corresponding filling machines, the operation time and the manual operations in connection with the filling process are reduced. This implies that a high filling capacity combined with a minimum number of operators can be achieved.

Each filling machine on the carrousel has its own connections to LPG, compressed air and electrical power via the carrousel. This implies that the supply of LPG, compressed air and electrical power can be disconnected on each individual filling machine, without taking the whole carrousel out of service. This makes it possible, in case of repair or periodic maintenance of the filling machines, to take one or more machines out of service, without disturbing the remaining production.

In order to make maximum use of the high production capacity of the filling carrousel it can be supplied with equipment for automatic introduction of cylinders from the chain conveyor to the filling machines on the carrousel, and with automatic ejection of cylinders from the filling machines back onto the chain conveyor. Moreover, if the filling machines of the carrousel are equipped with automatic or self-decoupling filling heads, one single operator will be able to supervise filling of up to 1600 cylinders per hour, thus depending on the filling time of the cylinders and the size of the carrousel.

Filling carrousels can be dimensioned so they are prepared for future increased capacity demands. At first, a carrousel can be equipped with the number of filling machines, which are necessary to comply with the capacity demand on a short view, e.g. 12 filling machines on a carrousel that can accommodate a total of 24 filling machines. Later on it is possible if desired or necessary, to increase the capacity by mounting another 12 filling machines on the existing carrousel.

The filling carrousel type EFC/U-24 consists of the following main components:

- A carrousel platform to accommodate a maximum of 24 filling machines, which is supported by self-aligning driving wheels placed on a running rail
- A central column that functions as axis of rotation for the carrousel platform, and as a distributor of LPG, compressed air and electrical power (via an incorporated collector ring unit) to each filling machine, with incorporated connection to LPG at the top of the central column and connections to compressed air and electrical power at the bottom of the central column (or connection to LPG at the bottom and connections to compressed air and electrical power at the top).

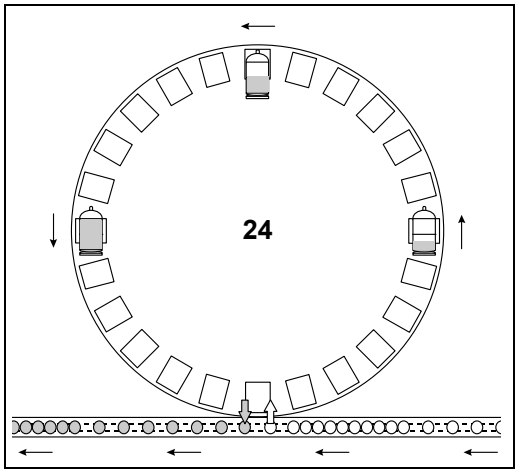
The filling carrousel can be supplied with clockwise or anti-clockwise rotation direction.

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Advantages

- Construction designed for an industrial environment
- The filling carrousel is easily maintained as all the components are easily accessible and easy to maintain, service or replace
- High filling capacity combined with a minimum number of operators
- The design with self-aligning driving wheels and a running rail combined with a very solid construction, gives a steady and balanced rotation, which is extremely important in order to achieve a high filling accuracy on the filling machines. The self-aligning driving wheels ensure that the wear of the running rail is reduced to a minimum.
- Possibility of, via a hydraulic driving unit, adjusting the rotation speed of the carrousel steplessly, according to the filling time of the cylinders, which ensures a maximum utilization of the capacity, as well as possibility of manual operation during functional test or service
- Possibility of repairing or servicing one or more filling machines on the carrousel without disturbing the remaining production
- Possibility of mounting the number of filling machines which are necessary to achieve the present capacity demand, and at the same time prepare the carrousel for future increased capacity demands
- Possibility of stepwise system upgrading through easy and fast connection to peripherical equipment, such as a communication module with corresponding PC and a central tare encoding station

Functional description



The cylinders are transferred from the chain conveyor to a free filling machine on the carrousel by an introduction unit. When the cylinder is placed correctly on the filling machine, it is filled up to a pre-set weight. After a completed filling the cylinder is transferred back onto the chain conveyor by an ejection unit.

In order to utilize the carrousel optimally, the rotation speed of the carrousel must be adjusted with accuracy according to the filling time of the cylinders, to ensure that a cylinder undergoes a complete filling process during one carrousel rotation.

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Scope of supply

The filling carrousel is supplied for stationary installation on floor, including corresponding carrousel platform accommodating a maximum of 24 filling machines, self-aligning driving wheels, running rail, and central column for connection to LPG at the top of the central column, and connections to compressed air and electrical power at the bottom of the central column (or connection to LPG at the bottom and connections to compressed air and electrical power at the top).

Additional equipment

(Only included if specified in quotation)

- Hydraulic driving unit to rotate the carrousel, with incorporated stepless speed regulator, which is used for accurate adjustment of the rotation speed of the carrousel according to the filling time of the cylinders
- Introduction unit for automatic introduction of cylinders to the filling machines on the carrousel from the chain conveyor
- Ejection unit for automatic ejection of cylinders from the filling machines on the carrousel to the chain conveyor
- Power supply (12 V DC) in flameproof enclosure (Ex d) to be placed in hazardous area at the carrousel system including cabling to the filling machines for supply of max. 24 filling machines.
- Communication module and PC including software for data collection
- Central tare encoding station
- LPG filling machines
- Chain conveyor plant
- Pressure regulation arrangement with filter, differential pressure valve and gas stop valves in order to maintain a desired constant filling pressure on the filling system
- Valve orientation unit for correct orientation of the cylinders before they are introduced to the filling machines on the carrousel from the chain conveyor (only in connection with filling of cylinders with screw valves)
- Standard spare parts set

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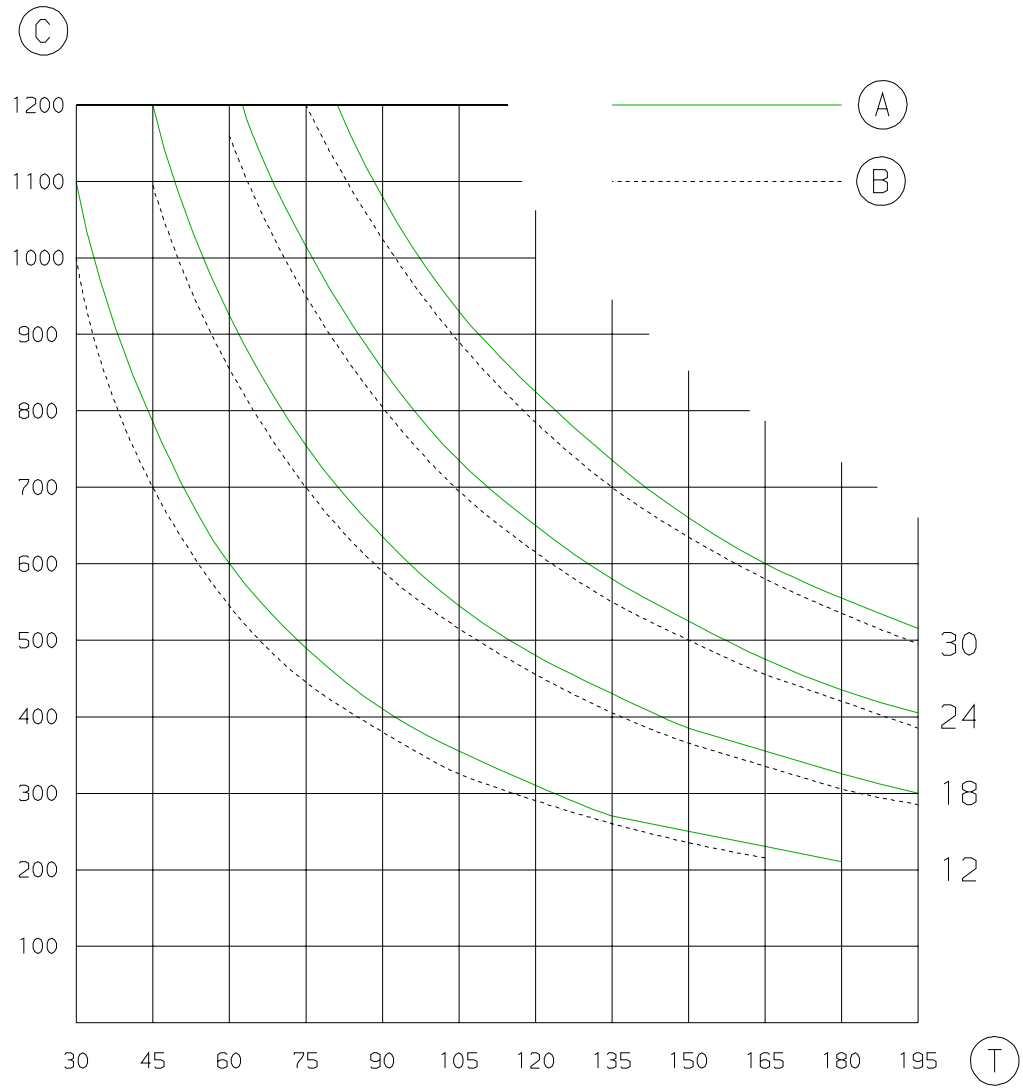
Technical data

Capacity and conditions	• See “Capacity diagram”
Necessary connections (specifications and quality requirements)	• LPG installation (LPG to the filling machines): – Connection size: 3” pipe – Filling pressure: Max. 2.1 MPa. (21 bar) – Test pressure: Max. 3.0 MPa. (30 bar) • Compressed air installation (compressed air to the filling machines): – Connection size: G1/2” BSP – Working pressure: Min. 0.6 MPa. (6 bar) - Max. 1.0 MPa. (10 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2 [1)] • Electrical power installation (electrical power to the filling machines via 12 V DC power supply): – Voltage: 1-phase + neutral + earth (voltage variation from 90 to 264 V AC) – Frequency: Variation from 48 to 62 Hz • Earth net: – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm
Consumption data	• The filling carrousel itself does not consume compressed air nor electrical power. The carrousel only functions as a distributor of compressed air and electrical power to each individual filling machine via the central column.
Weight	• Approx. 4900 kg (including 24 filling machines type UFM)
Static surface load	• < 0.29 N/mm² (3.0 kp/cm²)
Approvals	• The filling carrousel is designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500
Other data	• Diameter: 5800 mm (including filling machines type UFM) • Level: 550 mm • Rotation time: Min. 38 seconds/rotation - Max. 470 seconds/rotation • Central column: Connection to LPG at the top of the central column and connections to compressed air and electrical power at the bottom of the central column (or connection to LPG at the bottom and connections to compressed air and electrical power at the top)

[1)] The compressed air must not contain particles larger than 1 μm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.

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Capacity diagram



C Capacity (cylinders/hour)

T Filling time (seconds/ cylinder)

A Capacity curve for automatic filling of cylinders with centre valves

B Capacity curve for manual filling of cylinders with screw valves

12 Capacity curves for a filling carrousel type EFC/U-12 with 12 filling machines

18 Capacity curves for a filling carrousel type EFC/U-18 with 18 filling machines

24 Capacity curves for a filling carrousel type EFC/U-24 with 24 filling machines

30 Capacity curves for a filling carrousel type EFC/U-30 with 30 filling machines

The capacity curves shown in the diagram are only intended as a guide, and apply to filling carrousels connected to a chain conveyor plant via automatic introduction and ejection units.

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Dimensions in mm
Connection to LPG at the top of the central column and connections to compressed air and electrical power at the bottom

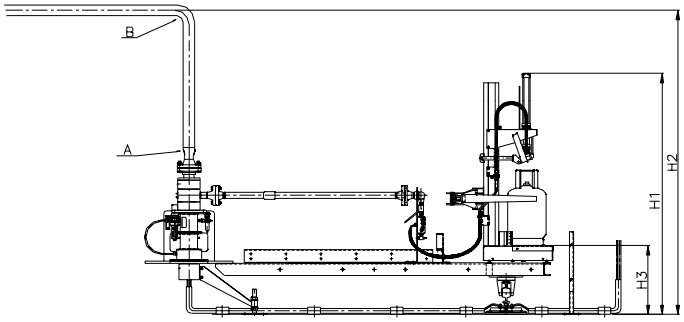
A = 2½" flange with reducer 2½"-3" (supplied by Kosan Crisplant)

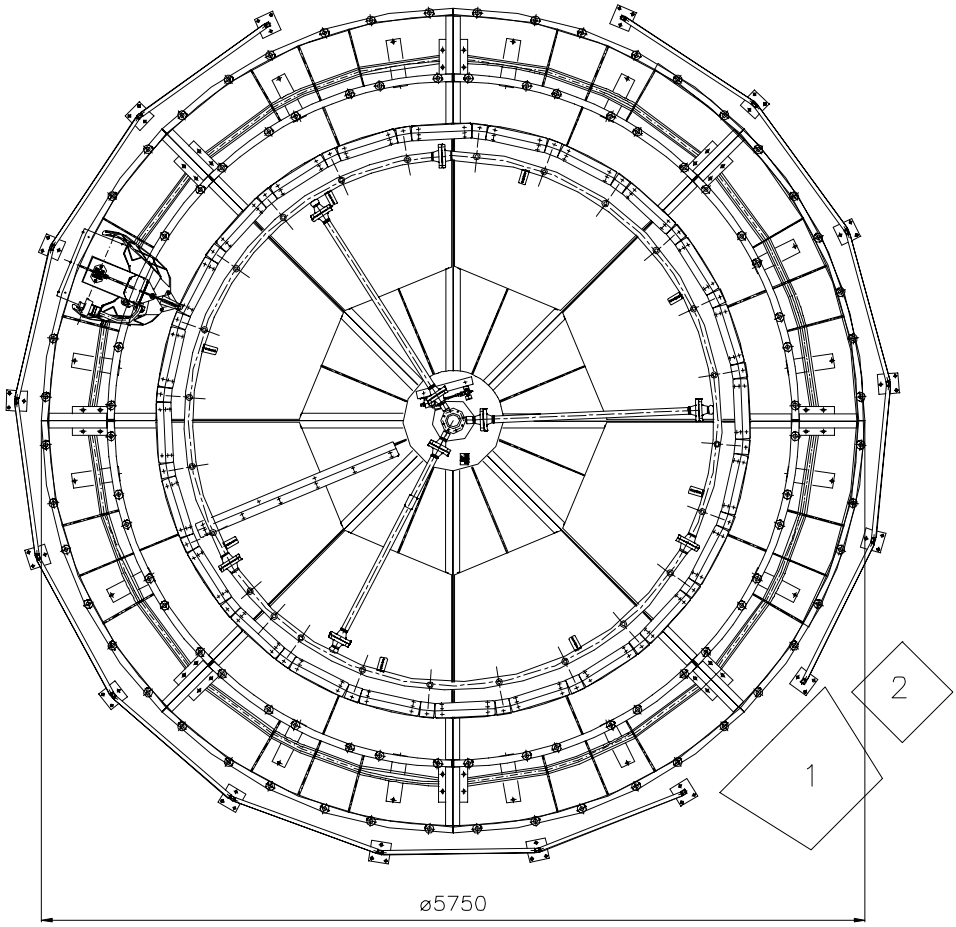
B = 3" LPG pipe (supplied by customer) to be welded on to reducer

H1 = Max. 3100 mm

H2 = H1+500 mm (recommended height)

H3 = level 550 mm





1 = Hydraulic driving unit
2 = Pump unit

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Dimensions in mm
Connection to LPG at the bottom of the central column and connections to compressed air and electrical power at the top

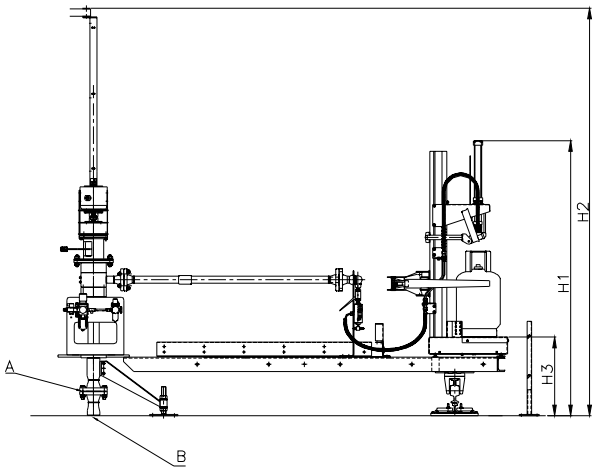
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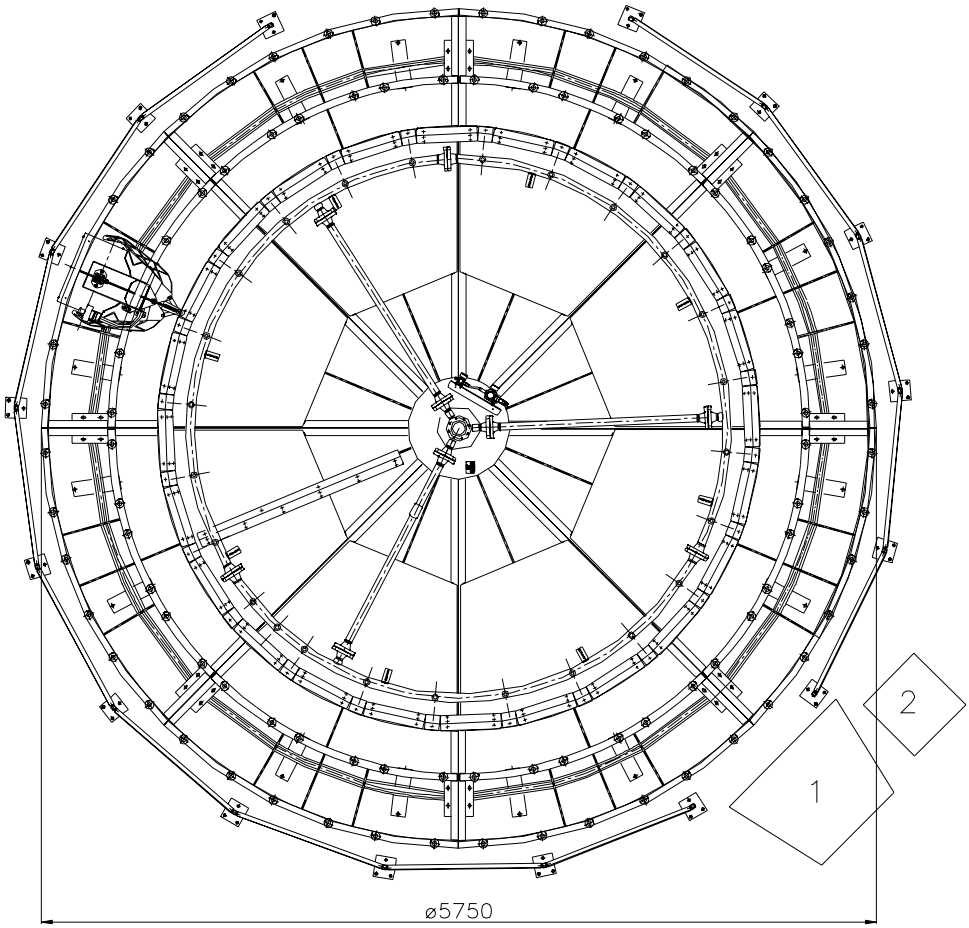
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